

Your Starter Pack to Agentic Automation [Tamil Edition]

Autonomous Agent — Use Case & Problem Statement

AI Academic Query Agent | Student & Academic Services Domain

Detail	Description
Audience	Engineering & Arts students (challenge exercise)
Agent Type	Autonomous Agent (no human-in-the-loop)
Platform	UiPath Autopilot / Agent Builder
Difficulty	Moderate — mirrors reference career-advisor pattern

1. Problem Statement

Rajalakshmi Engineering College receives a continuous stream of student emails and portal queries related to academic matters — attendance shortage, exam hall ticket issues, revaluation requests, internal marks clarification, and timetable changes. Currently, department office staff manually read every email, classify its intent, extract the relevant details, and decide whether the matter needs record-keeping, an automated reply, or specialist attention. This manual triage is repetitive, slow during peak periods (before exams or attendance freeze dates), and prone to inconsistent handling.

Design and build an Autonomous UiPath Agent, the AI Academic Query Agent, that reads an incoming student academic query, reasons over its content, classifies it, extracts structured details, and independently completes the required action end-to-end — without any human approval step — by invoking RPA workflows and Generative AI activities as tools.

2. System Prompt

You are an AI Academic Query Agent for Rajalakshmi Engineering College.
Your goal is to analyze student academic emails and decide the best possible action.

Follow this reasoning process:

1. Read and understand the email content.
2. Classify it into one of the following:
 - Attendance Shortage Query
 - Exam / Hall Ticket Query
 - Revaluation Request
 - Other
3. If it is Attendance Shortage or Exam/Hall Ticket or Revaluation:
 - Extract student register number, subject/course, and semester
 - Decide if it is urgent (e.g., deadline within 3 days)
 - If urgent, recommend immediate logging and notification
 - Generate a professional acknowledgement reply
4. If it is Other:
 - Skip processing

Rules:

- Think step-by-step before answering
- Be concise and structured
- Do not assume missing details

```
- Return output in structured format
```

3. User Prompt

```
Analyze the following student email and decide the appropriate action:  
{{email_text}}
```

4. Input Schema

```
{  
  "email_text": "string"  
}
```

5. Output Schema

```
{  
  "category": "string",  
  "register_number": "string",  
  "subject_course": "string",  
  "semester": "string",  
  "urgency": "string",  
  "action": "string"  
}
```

6. Tool: Build Workflow & Publish

Create the following process inputs for the RPA workflow the agent will invoke:

- category (text)
- register_number (text)
- subject_course (text)
- semester (text)
- urgency (text)

7. Add Tool: Map Inputs

Agent Field	Process Argument
category	in_category
register_number	in_register_number
subject_course	in_subject_course
semester	in_semester
urgency	in_urgency

8. Updated System Prompt — Tool Invocation Logic

If the email is classified as Attendance Shortage Query, Exam/Hall Ticket Query, or Revaluation Request:

- Use the `log_to_excel` tool to store the extracted student details in the academic query register.
- Use the `generate_reply` GenAI activity to draft a professional acknowledgement email to the student.
- If `urgency = "High"`, use the `notify_department` RPA workflow to alert the concerned department coordinator automatically (fully autonomous, no manual approval required).

If the email is classified as Other:

- Take no further action.

9. Tools & Activities Summary

Tool / Activity	Type
<code>log_to_excel</code>	RPA Workflow (UiPath Excel Automation)
<code>generate_reply</code>	GenAI Activity (LLM prompt-based text generation)
<code>notify_department</code>	RPA Workflow (Outlook / Teams notification)
<code>classify_query</code>	GenAI Activity (embedded in Agent reasoning)

Note: This use case is intentionally fully autonomous — the agent completes classification, extraction, logging, notification, and reply generation without an escalation or human-review step, demonstrating end-to-end autonomous decision-making for the workshop challenge.

10. Expected Learning Outcome

- Design a multi-step reasoning system prompt for an autonomous agent
- Define structured input/output schemas for agent-to-workflow communication
- Map agent-extracted fields to RPA process arguments
- Combine RPA workflows and GenAI activities as callable tools within one agent
- Publish and test a complete autonomous agent in UiPath Agent Builder